7.4 Inspection before trial operation

Inspection procedures before trial operation are provided in the following sections. Do not connect motor and mechanism while executing trial operation. If the motor cannot be removed from the mechanism, its load must be removed. The purpose of trial operation is to check the combination of the servo drive and motor as well as the wiring of servo drive. Perform inspection by referring to the inspection procedure of the motor in use.

7.4.1 Inspection procedure for servo motor (AC)

While using HIWIN EM1 series servo motor, perform inspection by referring to the procedure provided in Table 7.1.

Table 7.1: Inspection procedure for servo motor (AC)

Item	Description	Reference
Hardware	1 Check if the servo drive is correctly installed inside the control box.	Refer to section 4.1.2
	2 Check the wiring of the servo drive. – CN1 power-Check the voltage of the input power. Check if the connectors are firmly connected. – CN2 motor power-Check if the UVW power terminals on the servo drive and motor power cable are correctly wired. Check if the terminals are secure. – CN7 encoder-Check if the motor and the servo drive are firmly connected.	
	3 Confirm the encoder information. Ensure the software setting is correct.	
	4 Loosen the coupling. Do not connect the motor and mechanism.	
Software	5 Download the latest version of Thunder. And connect to the servo drive.	Refer to section 7.2
	6 Do software setting by following the procedures provided in Thunder.	Refer to section 7.3
	7 Check the moving direction. Execute trial operation, such as JOG or point-to-point (P2P) motion.	Refer to section 0
	8 Operate with controller.	Refer to section 10.1

7.4.2 Inspection procedure for other motor

While using customized servo motor, linear motor, direct drive motor or torque motor, detection for electrical angle must be completed before operation. The combinations of motors and encoder signals are provided in Table 7.2.

Table 7.2: Combinations of customized AC/LM/DM/TM and encoder signals

Motor	Encoder Signal	Excellent Smart Cube (ESC)
Customized servo motor	Tamagawa 2.5 MHz	Not required
Linear motor	Digital TTL signal	Not required
Linear motor	Digital TTL signal + digital Hall sensor signal	Required (ESC-SS)
HIWIN direct drive motor with absolute feedback system	Absolute serial signal	Not required
Linear motor, direct drive motor with incremental feedback system or torque motor	Analog sin/cos signal	Required (ESC-AN)
Linear motor and torque motor	Serial EnDat or BiSS-C signal	Required (ESC-SS)
Linear motor, direct drive motor or torque motor	Analog sin/cos signal + digital Hall sensor signal	Required (ESC-AN)

Table 7.3: Inspection procedure for customized AC/LM/DM/TM

Item	Description	Reference
Hardware	1 Check if the servo drive is correctly installed inside the control box.	Refer to section 4.1.2
	2 Check the wiring of the servo drive. ▶ CN1 power-Check the voltage of the input power. Check if the connectors are firmly connected. ▶ CN2 motor power-Check if the UVW power terminals on the servo drive and motor power cable are correctly wired. Check if the terminals are secure. ▶ CN7 encoder-Check if the motor and the servo drive are firmly connected. If Hall sensor is installed, check if the wiring and connectors are secure.	
	3 Confirm the encoder information. Ensure the software setting is correct.	
	4 Loosen the coupling. Do not connect the motor and mechanism.	
Software	5 Download the latest version of Thunder. And connect to the servo drive.	Refer to section 7.2
	6 Do software setting by following the procedures provided in Thunder.	Refer to section 7.3
	7 Check the moving direction. Complete detection for electrical angle.	Refer to section 7.5
	8 Execute trial operation, such as JOG or point-to-point (P2P) motion.	Refer to section 0
	9 Operate with controller.	Refer to section 10.1

7.5 Detection for electrical angle

While using customized servo motor (AC), linear motor (LM), direct drive motor (DM) with incremental feedback system or torque motor (TM), detection for electrical angle must be completed before closed loop control. ED1 series servo drive provides three detection methods: SW method 1, STABS test/tune, Digital hall and Analog Hall.

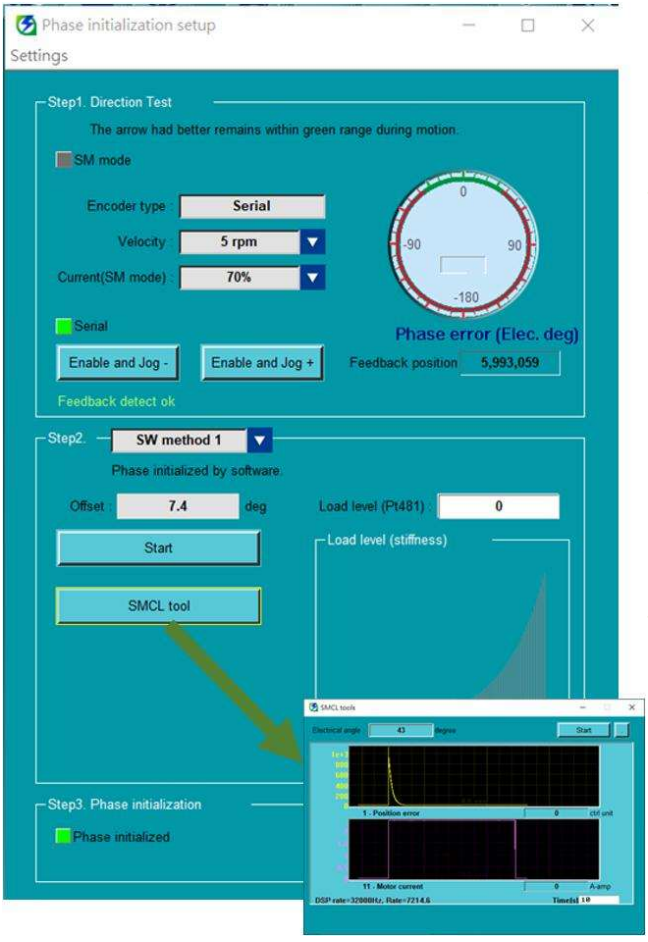
7.5.1 SW method 1

While detecting electrical angle by SW method 1, refer to Table 7.4 for applicable combinations of motors and encoder signals.

Table 7.4: Applicable combinations for SW method 1

Motor	Encoder Signal	Excellent Smart Cube (ESC)
Customized servo motor	Tamagawa 2.5 MHz	Not required
Linear motor	Digital TTL signal	Not required
Linear motor, direct drive motor or torque motor	Analog sin/cos signal	Required (ESC-AN)

Fig. 7.2: Operating procedure of SW method 1



1. Select velocity and current for detecting electrical angle. Click on **Enable and Jog+** and **Enable and Jog-** buttons to move the motor. While the motor is moving, check if the electrical angle falls in the range coloured in green.
2. Select **SW method 1** and click on **Start** button for three times. The difference of offset must not exceed 5 deg.

   Example:

   Offset: 73.5 deg

   Offset: 74.1 deg

   Offset: 72.3 deg

   Open **SMCL tool** and observe position deviation during execution. If the position deviation is not close to 0 within one second, it means the gain is improper, please adjust load level.
3. Click on **Start phase initialization** button. Wait till detection for electrical angle completes and check **Phase initialized** indicator. If **Phase initialized** indicator is green, it means electrical angle has been successfully detected.

Note:

- If SW method 1 is executed under open loop control, the motor will be automatically disabled to avoid overheating when it stops for a period of time.
- If the load level is too high, it may cause mechanical resonance.
- (3) If the motor vibrates during the execution of SW metohd1, the user can adjust Pt489 and Pt48A until the vibration stops. Then SMCL tool can be used to confirm that the convergence is good, and the user can go to step 3.